DRIFTLESS — Feature-to-Code Mapping & Cooperation Analysis

1. Master Feature-to-Code Traceability Matrix

Feature	Frontend Files	Extension Files	Backend Files	API & Database Dependencies	Implementation Status
User Registration & Persona Onboarding	`dashboard/core/templates/auth.html`	None	`core/views.py:73`, `core/serializers.py:63`, `core/models.py:16`	`POST /api/auth/register/`, `User`, `UserProfile`, `UserSettings`, `SiteLimit`	**PASS (Fully Verified)**
Multi-Identifier Login (Username/Email/Phone)	`dashboard/core/templates/auth.html`	None	`core/views.py:92`, `core/models.py:16`	`POST /api/auth/login/`, `Token`, `User`, `UserProfile`	**PASS (Fully Verified)**
API Token Copy & Regeneration	`dashboard/core/templates/dashboard.html`	`extension/popup.html`, `extension/popup.js`	`core/views.py:150`	`POST /api/auth/regenerate-token/`, `authtoken_token`	**PASS (Fully Verified)**
Passive Tab Dwell Tracking	None	`extension/background.js`	None (Local accumulation)	`chrome.tabs`, `chrome.windows`, `chrome.idle`, `chrome.storage.local`	**PASS (Fully Verified)**
Automatic Idle Detection	None	`extension/background.js:208`	None	`chrome.idle.setDetectionInterval(60)`	**PASS (Fully Verified)**

Feature	Frontend Files	Extension Files	Backend Files	API & Database Dependencies	Implementation Status
Periodic 60s Delta Sync	None	`extension/background.js:263`, `extension/manifest.json`	`core/views.py:165`, `core/serializers.py:116`, `core/models.py:214`	`POST /api/usage/`, `UsageLog`, `chrome.alarms`	**PASS (Fully Verified)**
Midnight Dwell Commit & Reset	None	`extension/background.js:217`	`core/views.py:165`	`POST /api/usage/`, `chrome.alarms`, `declarativeNetRequest`	**PASS (Fully Verified)**
3-Tier Escalation: 80% Nudge	None	`extension/background.js:93`, `extension/popup.js:133`	None	`chrome.notifications.create`, `chrome.storage.local`	**PASS (Fully Verified)**
3-Tier Escalation: 100% Alert	None	`extension/background.js:101`	None	`chrome.notifications.create`, `chrome.storage.local`	**PASS (Fully Verified)**
3-Tier Escalation: 5m Grace + DNR Block	None	`extension/background.js:109`, `extension/blocked.html`	None	`chrome.declarativeNetRequest.updateDynamicRules`	**PASS (Fully Verified)**
Focus Mode Toggling	`dashboard/core/templates/dashboard.html:1115`	`extension/popup.js:201`, `extension/background.js:352`	`core/views.py:443`, `core/models.py:246`	`PATCH /api/settings/`, `UserSettings.focus_mode_active`	**PASS (Fully Verified)**
Timed Focus Sprints (25/45/60 m)	None	`extension/popup.js:228`, `extension/background.js:416`	`core/views.py:577`, `core/models.py:199`	`POST /api/sessions/`, `FocusSession`, `chrome.alarms`	**PASS (Fully Verified)**

Feature	Frontend Files	Extension Files	Backend Files	API & Database Dependencies	Implementation Status
Per-Site Daily Limit Configuration	`dashboard/core/templates/dashboard.html:1212`	`extension/popup.js:245`	`core/views.py:467`, `core/serializers.py:135`, `core/models.py:174`	`GET/POST/PATCH/DELETE /api/limits/`, `SiteLimit`	**PASS (Fully Verified)**
Dynamic Domain Categorization	None	`extension/popup.js:141`	`core/models.py:106` (`classify_domain`)	`CATEGORY_MAP`, `UserProfile.persona`, academic domain regex	**PASS (Fully Verified)**
User Category Overrides	`dashboard/core/templates/dashboard.html:1261`	None	`core/views.py:526`, `core/serializers.py:152`, `core/models.py:153`	`GET/POST/DELETE /api/categories/`, `CategoryOverride`	**PASS (Fully Verified)**
7-Day Analytics & Charting	`dashboard/core/templates/dashboard.html:913`	None	`core/views.py:213`	`GET /api/summary/`, `UsageLog`, Chart.js 4.4	**PASS (Fully Verified)**
Strict Streak Calculation	`dashboard/core/templates/dashboard.html:874`	`extension/popup.js:61`	`core/views.py:294`	`GET /api/summary/`, `UsageLog`, `SiteLimit`, `UserSettings`	**PASS (Fully Verified)**
Focus Goal Ring (3h Default)	`dashboard/core/templates/dashboard.html:890`	None	`core/views.py:213`, `core/models.py:253`	`UserSettings.focus_goal_minutes`, SVG `stroke-dasharray`	**PASS (Fully Verified)**
Dual-Theme System (Calm Sky / Opal Noir)	`dashboard/core/templates/*.html`, `core/static/css/theme.css`	`extension/popup.html`, `extension/blocked.html`, `theme.css`	None	CSS custom properties, `localStorage['driftless-theme']`	**PASS (Fully Verified)**
Demo vs. Live Data Switcher	`dashboard/core/templates/dashboard.html:786`	None	`core/management/commands/seed_demo.py`	Client toggle switching between fallback demo payload and `/api/summary/`	**PASS (Fully Verified)**

Feature	Frontend Files	Extension Files	Backend Files	API & Database Dependencies	Implementation Status
1-Click Data Export (JSON)	`dashboard/core/templates/dashboard.html:1285`	None	`core/views.py:603`	`GET /api/data/export/`, `UsageLog`, `SiteLimit`, `FocusSession`	**PASS (Fully Verified)**
1-Click Browsing History Erasure	`dashboard/core/templates/dashboard.html:1289`	`extension/popup.js:318`	`core/views.py:632`	`POST /api/data/delete/`, `RESET_ALL_TIMERS` message	**PASS (Fully Verified)**

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2. Multi-File Component Cooperation Analysis

2.1. Cooperating Modules: Per-Site Limits Enforcement

A user configures a 25-minute limit on `reddit.com`. Here is how the files cooperate:
1. **Creation (`dashboard.html` or `popup.js`):** The user enters the domain and limit. An asynchronous `POST /api/limits/` is dispatched to the Django backend.
2. **Persistence (`core/views.py` & `core/models.py`):** The DRF view validates `1 ≤ daily_limit_minutes ≤ 1440` and updates the `SiteLimit` record in `db.sqlite3`.
3. **Synchronization (`background.js`):** During the next 60-second pull cycle, `pullBackendConfig()` requests `GET /api/limits/` and caches the limit in `chrome.storage.local.set({ siteLimits })`.
4. **Enforcement (`background.js`):** As the user spends time on `reddit.com`, `evaluateSiteLimits()` tracks accumulated seconds against 25 minutes. At 20 minutes (80%), a desktop nudge is created. At 25 minutes (100%), an alert is shown. At 30 minutes (100% + 5m), a DNR redirect rule is created.
5. **Interception (`blocked.html`):** Future requests to `reddit.com` are caught by Chrome's network engine and redirected to `chrome-extension:// ... /blocked.html?site=reddit.com`.
6. **Analytics (`views.py` & `dashboard.html`):** The dashboard `/api/summary/` checks `reddit.com` against the `SiteLimit`. If exceeded, it breaks the consecutive daily streak.

2.2. Cooperating Modules: Persona-Driven Onboarding

1. **Registration UI (`auth.html`):** The user registers, choosing persona `"influencer"`.
2. **Validation & Defaults (`core/serializers.py`):** `RegisterSerializer` calls `apply_persona_defaults()`, seeding initial `SiteLimit` records (e.g. Netflix: 30m, Twitch: 25m,

Reddit: 20m).

3. **Classification (`core/models.py`):** `classify_domain()` checks `user.profile.persona`. When the influencer browses `instagram.com` or `youtube.com`, the engine categorizes it as `Productivity` instead of `Social` or `Entertainment`.
4. **Visualization (`dashboard.html`):** Time spent on Instagram is rendered in emerald green on the 7-day chart, increasing the user's Focus Efficiency score.

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3. Implementation State Classifications

- **Verified Live Full-Stack Features:**
- Multi-identifier authentication, Token auth, persona seeding.
- Periodic delta ingestion, SQLite persistence, and aggregation.
- Per-site limits, category overrides, and data export/deletion.
- Active tab tracking, 3-tier escalation, DNR blocking, and focus sprint timers.
- Calm Sky & Opal Noir dual-theme styling across dashboard, popup, and blocked screen.
- **Client-Side Isolated Features:**
- Timer countdown ticker (`popup.js` `setInterval`).
- Active window focus and 60-second idle detection (`chrome.idle`).
- FOUC-prevention scripts in HTML `<head>`.
- **Demo Mode Fallback:**
- The dashboard includes a built-in toggle (`Demo` vs. `Live`). In Demo mode, it renders a 14-day sample payload without requiring an active server connection. In Live mode, it fetches directly from `/api/summary/`.
